

PROJECT DOCUMENTATION

Operating Room Efficiency Investigation

Analysis of surgical delays, operating room workload, scheduling performance, and resource utilization

Project at a glance

2,172 surgical encounters • 8 OR suites • 10 surgical services • January 3, 2022 to March 31, 2022

Tools: PostgreSQL 18 • SQL • Power BI • Excel/CSV

Project type: Healthcare operations / Data Analytics case study

Purpose of this document

This document explains the project from the business question through data validation, SQL analysis, KPI selection, Power BI development, and final findings. It is written so a recruiter can quickly understand what was done, why it was done, and what the analysis can realistically tell us.

1. Executive Summary

This project analyzes surgical encounter data from January 3, 2022 to March 31, 2022 to understand operating room workload and scheduling patterns. The focus was not only on how many surgeries were performed, but also on how much time ORs were occupied and how much scheduling delay occurred.

The data was loaded into PostgreSQL, validated, and prepared with three derived measures: procedure duration, OR occupancy time, and scheduling delay. SQL was then used to answer business questions across OR suites, services, procedures, and time patterns. The results were presented in a three-page Power BI dashboard.

Main conclusion

OR workload is better understood through volume, occupancy, scheduling delay, and procedure mix together. Looking only at surgery count can give an incomplete view of workload.

2. Business Problem

An OR can handle many short procedures or fewer long procedures. Because of this, surgery count alone does not show how much time an OR was actually occupied. Scheduling delay adds another operational dimension that is also not visible from case volume.

The core business question

Where are the main differences in OR volume, occupied time, and scheduling delay, and what do those differences tell us about OR workload?

3. Why This Analysis Matters

The analysis separates three measures because each answers a different operational question.

Measure	Question it answers	Why it matters
Surgery volume	How many cases were performed?	Shows where case activity is concentrated.
OR occupancy	How much time was the OR occupied?	Adds a workload view that case count cannot provide by itself.
Scheduling delay	How large was the gap between scheduled time and actual start?	Shows where scheduling timing differs across ORs.

No combined efficiency score was created. Volume, occupancy and delay describe different parts of the OR process.

4. Project Objectives

- Understand overall surgical workload.
- Compare surgery volume across OR suites.
- Compare workload and occupancy across surgical services.
- Identify frequent procedures and longer-duration procedures.

- Measure scheduling delay across ORs and services.
- Compare booked OR time with actual OR occupancy.
- Review daily and weekly surgery patterns.
- Build a Power BI dashboard that clearly presents the findings.

5. Project Contribution

The contribution was the complete analytical workflow, from raw data to a business-facing dashboard.

Stage	What was done
Data preparation	Loaded the encounter data into PostgreSQL and created analytical fields from timestamps.
Data validation	Checked completeness, duplicate IDs, booked-time validity, and timestamp order.
SQL analysis	Performed business-focused analyses for volume, occupancy, delay, procedures, services, and booked-vs-actual time.
KPI identification	Selected measures that directly reflect the available data.
Power BI	Built a three-page dashboard for overall, OR/service, and insight views.
Business interpretation	Translated results into findings and review areas without claiming unsupported root causes.

6. Dataset Understanding

The source dataset is OR Utilization(3).csv. The working PostgreSQL table is public.or_utilization. The data grain is one row per surgical encounter, with encounter_id as the business identifier.

Dataset detail	Value
Analysis period	January 3, 2022 – March 31, 2022
Surgical encounters	2,172
Original columns	12
OR suites	8
Surgical services	10
Business identifier	encounter_id
Data grain	One row = one surgical encounter

The original fields include encounter details, surgery date, OR suite, service, CPT information, booked time, scheduled time, and operational timestamps.

7. Data Preparation & Validation

Derived field	Calculation	Meaning
Procedure Duration	End Time – Start Time	Actual procedure time.
OR Occupancy	Wheels Out – Wheels In	Total time the OR was occupied by the encounter.
Scheduling Delay	Start Time – OR Schedule	Time between scheduled OR time and actual procedure start.

Validation checks included:

- Row and column counts.
- Duplicate encounter IDs.
- Missing values.
- Non-positive booked time.
- Logical ordering of OR timestamps.

Validation result

The working data passed the main checks. No duplicate encounter IDs, missing values, or invalid booked-time values were identified, and the expected timestamp sequence was valid.

8. SQL Business Analysis

The SQL work was organized into `data_preparation.sql`, `data_validation.sql`, and `EDA_OR_utilization_analysis.sql`. The EDA script contains 13 business analyses.

The analyses were built around practical questions:

- How many surgeries were performed overall and by OR?
- Which ORs had the highest total occupancy?
- Which ORs had the highest average scheduling delay?
- Which procedures were most frequent and which had longer average duration?
- How does booked time compare with actual OR occupancy?
- Which services account for the most OR occupancy?
- How does scheduling delay vary by service?
- How do ORs compare when volume, occupancy and delay are viewed together?

Why the OR performance summary mattered

The final OR performance analysis brought surgery count, occupancy hours and average scheduling delay into one result, making it easier to see why case volume alone is not enough.

9. KPI Identification

KPI	Result	What it represents
Total Surgeries	2,172	Total surgical encounters.
Average Daily Surgeries	35.03	Average surgeries per day.

Average OR Occupancy	79.70 min	Average actual OR occupancy per encounter.
Total OR Occupancy	≈ 2.89K hours	Total accumulated OR occupancy.
Average Procedure Duration	45.48 min	Average procedure time.
Average Scheduling Delay	57.07 min	Average gap between scheduled time and actual start.
Booked vs Actual OR Time Variance	+2.51 min	Difference between average booked time and actual OR occupancy.

10. Key Business Findings

OR 3 had the highest surgery volume.

OR 3 recorded 439 surgeries, while OR 8 recorded 186.

OR occupancy tells a different story from case count.

OR 1 had the highest total occupancy at 386.75 hours. OR 8 had the lowest case volume but still recorded 377.90 hours.

Scheduling delay varied across ORs.

OR 2 had the highest average delay at 94.48 minutes; OR 3 had the lowest at 21.85 minutes.

Orthopedics had the highest service-level occupancy.

Orthopedics recorded about 540.13 hours of occupancy and the highest service-level average scheduling delay at 89.01 minutes.

Ophthalmology had the highest service volume.

Ophthalmology recorded 334 surgeries; its occupancy was about 199.68 hours and average delay about 7.15 minutes.

Procedure mix also matters.

Extracapsular cataract removal was the most frequent procedure with 334 cases. Liposuction had the longest average duration at about 122 minutes.

Daily volume was relatively stable.

Average daily volume was 35.03 surgeries, with daily volume ranging from 32 to 42.

Tuesday had the highest surgery volume.

Tuesday recorded 456 surgeries; Monday recorded 388.

Booked and actual OR time were close on average.

Average booked time was 77.19 minutes and average actual occupancy was 79.70 minutes, a +2.51 minute variance.

11. What the Findings Mean

The main pattern across the project is that different measures answer different operational questions. OR 3 handled the highest number of surgeries and also had the lowest average scheduling delay. OR 8 handled the fewest surgeries but still had relatively high total occupancy.

For operational review, volume should therefore be read together with occupied time and scheduling delay. Orthopedics also stands out for further review because of its high occupancy and scheduling delay. The data identifies these patterns but does not explain their causes.

12. Power BI Dashboard

Page	Purpose	Main content
Page 1 – Executive Overview	Quick overall picture	Core KPIs, daily surgery volume, OR occupancy, procedure duration, and overall workload measures.
Page 2 – OR & Specialty Performance	Detailed comparison	OR and service-level volume, occupancy, scheduling delay, and related comparisons.
Page 3 – Key Insights & Recommendations	Decision-focused summary	Main findings, management considerations, and analytical limitation.

The dashboard follows the same logic as the SQL analysis: volume, occupancy and delay are shown as separate measures.

13. Management Considerations

The findings can be used as starting points for operational review. They are not presented as proof of root causes.

- Review ORs with higher scheduling delays, particularly OR 2.
- Look at occupancy as well as case volume when assessing workload.
- Review Orthopedics workload and scheduling patterns.
- Consider procedure mix when reviewing OR workload.
- Compare booked time with actual OR occupancy as part of scheduling review.
- Use daily and weekly volume patterns as supporting information when reviewing OR schedules.

Important interpretation point

The dataset does not contain enough information to say that staffing, equipment, surgeon behavior, patient complexity, cleaning, cancellations, or another specific factor caused the observed patterns.

14. Analytical Limitations

Why utilization % is not reported

Actual OR capacity hours are not available. The dataset tells us how much time was occupied, but not how many hours were available. Therefore, a true OR Utilization % cannot be calculated.

- Reasons for scheduling delays are not included.
- The available fields do not support causal explanations for differences between ORs or services.
- The analysis covers the available encounter records from January 3 to March 31, 2022.
- Booked time, procedure duration, OR occupancy, and scheduling delay are different measures and should not be treated as interchangeable.

15. Project Outcome

The project turned raw surgical encounter data into a structured analytical workflow. PostgreSQL was used for preparation and validation, SQL was used to answer business questions, and Power BI was used to present the results.

The final analysis gives a clearer view of where surgical activity is concentrated, which ORs account for more occupied time, where scheduling delay is higher, how services differ, and how booked time compares with actual occupancy.

Portfolio value

The project demonstrates an end-to-end approach: understand the business problem, validate the data, build SQL analysis, select useful KPIs, create a dashboard, and explain the findings in business terms.

16. Skills Demonstrated

Technical	Analytical / business
PostgreSQL and SQL data preparation	Healthcare operations analysis
Data-quality validation	KPI identification and interpretation
Exploratory data analysis	Time-based and OR/service comparison
Power BI dashboard development	Business insight generation
Analytical documentation	Communicating findings and limitations

17. End-to-End Project Flow

Step	What was done
1. Understand	Reviewed encounter-level data and operational timestamps.
2. Validate	Checked completeness, uniqueness, booked time and timestamp order.
3. Prepare	Created procedure duration, OR occupancy and scheduling delay.
4. Analyze	Ran 13 SQL business analyses.
5. Identify KPIs	Selected measures supported by the available data.
6. Visualize	Built a three-page Power BI dashboard.

7. Interpret	Summarized findings and management review areas.
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18. Conclusion

This project shows that OR workload is not a single number. Surgery volume, occupied time, scheduling delay, and procedure mix each add a different part of the story.

The strongest takeaway is simple: an OR with fewer cases can still have a high workload, and an OR with many cases can have a different scheduling pattern. Looking at these measures together gives a more useful starting point for operational review.

The SQL scripts, Power BI dashboard, documentation, and presentation are designed to tell the same story from different angles and form a complete portfolio case study.